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Original article

PKM2-dependent glycolysis promotes skeletal muscle cell pyroptosis by activating the NLRP3 inflammasome in dermatomyositis/polymyositis

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Abstract

Objectives. Muscle cell necrosis is the most common pathological manifestation of idiopathic inflammatory myopathies. Evidence suggests that glycolysis might participate in it. However, the mechanism is unclear. This study aimed to determine the role of glycolysis in the muscle damage that occurs in DM/PM.

Methods. Mass spectrometry was performed on muscle lesions from DM/PM and control subjects. The expression levels of pyruvate kinase isozyme M2 (PKM2), the nucleotide-binding and oligomerization domain-like receptor family pyrin domain containing 3 (NLRP3) inflammasome and pyroptosis-related genes in muscle tissues or plasma were determined by real-time PCR, western blot analysis, IF and ELISA. In addition, IFNγ was used to stimulate myotubes, and the relationships among PMK2 expression, NLRP3 inflammasome activation and pyroptosis were investigated.

Results. Mass spectrometry and bioinformatics analysis suggested that multiple glycolysis processes, the NLRP3 inflammasome and programmed cell death pathway-related proteins were dysregulated in the muscle tissues of DM/PM. PKM2 and the NLRP3 inflammasome were upregulated and positively correlated in the muscle fibres of DM/PM. Moreover, the pyroptosis-related proteins were increased in muscle tissues of DM/PM and were further increased in PM. The levels of PKM2 in muscle tissues and IL-1β in plasma were high in patients with anti-signal recognition particle autoantibody expression. The pharmacological inhibition of PKM2 in IFNγ-stimulated myotubes attenuated NLRP3 inflammasome activation and subsequently inhibited pyroptosis.

Conclusion. Our study revealed upregulated glycolysis in the lesioned muscle tissues of DM/PM, which activated the NLRP3 inflammasome and leaded to pyroptosis in muscle cells. The levels of PKM2 and IL-1β were high in patients with anti-signal recognition particle autoantibody expression. These proteins might be used as new biomarkers for muscle damage.

CLINICAL SCIENCE

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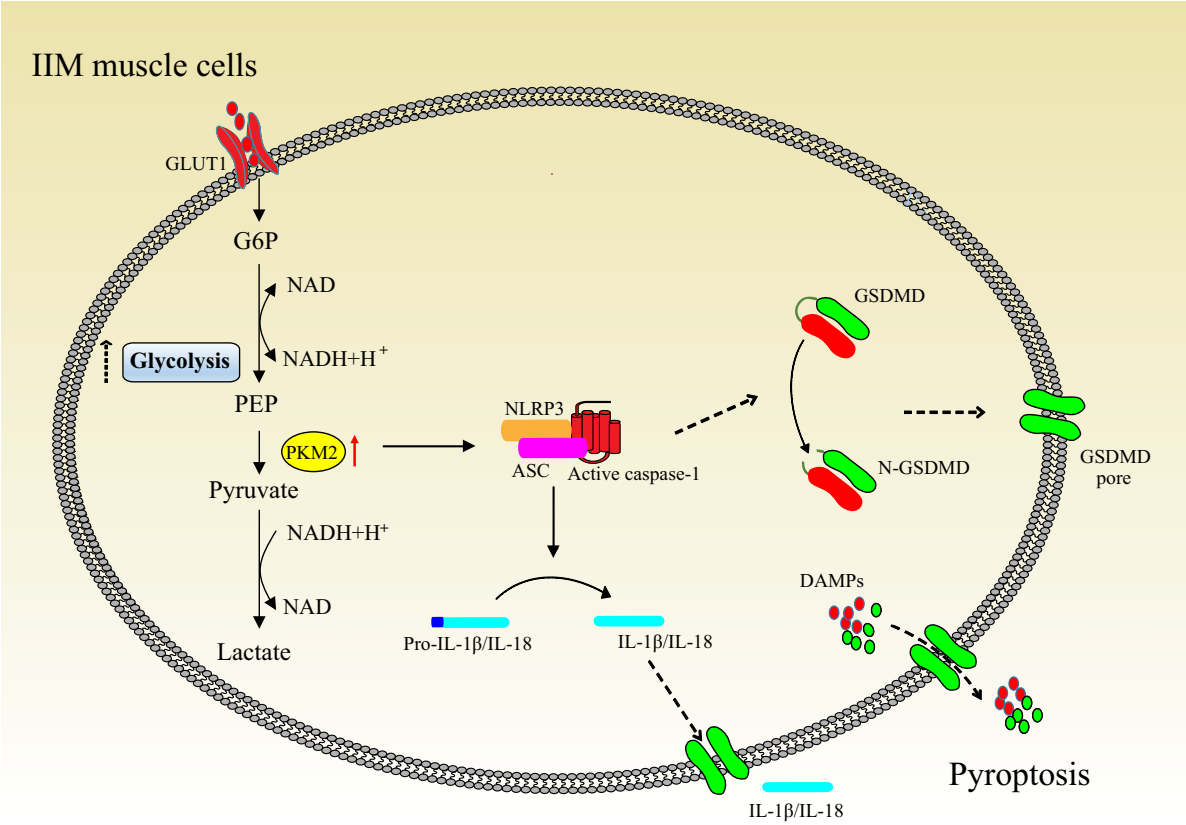
Graphical Abstract



Key words: DM, PM, glycolysis, PKM2, NLRP3 inflammasome, pyroptosis

Rheumatology key messages

- Glycolysis process, NLRP3 inflammasome and programmed cell death pathway-related proteins are dysregulated in DM/PM muscles.
- Upregulated PKM2 in DM/PM muscles might lead to muscle cells pyroptosis through activating NLRP3 inflammasome.
- Glycolytic metabolism abnormalities and pyroptosis might play an important role during muscle damage in IIMs.

Introduction

Idiopathic inflammatory myopathies (IIMs) are a group of systemic autoimmune diseases characterized by muscle weakness, inflammatory cell infiltration in muscle and extramuscular manifestations [1]. IIMs mainly include DM, PM and sporadic IBM in adults and JDM in children [2]. Histopathologically, the infiltration of inflammatory cells that leads to muscle necrosis and atrophy is a main feature of DM and PM [3]. However, the extent of inflammatory cell infiltration in muscles does not always correlate with the severity of muscle fibre damage. Under some conditions, muscle cell death occurs even in the absence of inflammatory cells [4, 5] (Supplement Fig. S1, available at *Rheumatology* online). Emerging evidence suggests that endoplasmic reticulum stress, autophagy and other nonimmune mechanisms participate in skeletal muscle fibre damage in IIMs [1, 4, 6], but the exact mechanisms have not been clarified.

The emerging field of immunometabolism focuses on the functional links between metabolic reprogramming and the immune system, which provides an additional

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dimension for understanding immunity in health and disease [7]. Six major metabolic pathways are involved in metabolism, including glycolysis, fatty acid oxidation, fatty acid synthesis, the tricarboxylic acid cycle, amino acid metabolism and the pentose phosphate pathway [8]. Glycolysis is a 10-step metabolic pathway through which glucose is converted into pyruvate and 2 adenosine triphosphate molecules are produced [9]. Under abundant oxygen conditions, pyruvate fuels the tricarboxylic acid cycle and oxidative phosphorylation, whereas under hypoxic conditions, pyruvate is converted into lactate [10]. However, rapidly proliferating cells rely on glycolysis even in the presence of sufficient oxygen. The metabolic switch (aerobic glycolysis) influences cell proliferation, differentiation and function [11]. Skeletal muscle is the major site responsible for glucose uptake. In skeletal muscle, lots of energy-sensing molecules have been shown to sense variations in energy homeostasis and trigger changes in gene expression, especially under pathological conditions [12]. The expression of innate immune receptors on myocytes and immune cells, such as Toll- and nucleotide-binding and oligomerization domain-like receptors, can also regulate skeletal muscle metabolism by cell–cell communication [13–15]. 18-Fluorodeoxyglucose is a glucose analogue tracer that can be used to measure glucose uptake. In DM/PM patients, 18-fluorodeoxyglucose PET/CT scans show increased glucose uptake in muscle tissues, suggesting dysregulation of glucose metabolism in the muscles of patients with IIMs [16].

The nucleotide-binding and oligomerization domain-like receptor family pyrin domain containing 3 (NLRP3) inflammasome is a large, cytoplasmic, multiprotein complex composed of a cytosolic sensor, adaptor apoptosis-associated speck-like protein containing a caspase-recruiting domain and caspase-1 [17]. Once NLRP3 inflammasome is activated, caspase-1 regulates the cleavage of pro-IL-1β and pro-IL-18 into their mature forms. Caspase-1 can also cleave gasdermin D (GSDMD) to induce cell pyroptosis [18]. Reports suggested that IL-1β and IL-18 play important roles in the initiation and progression of IIMs [19, 20]. In addition, the protein expression of NLRP3 and caspase-1 were also shown to increase in the muscle tissues of DM and PM patients [21]. However, the exact mechanism is unknown.

Multiple exogenous and endogenous stimuli can activate the NLRP3 inflammasome, including glycolysis. Pyruvate kinase (PK), the major rate-limiting enzyme, catalyses the final reaction in glycolysis. There are four isoforms of PK: the L (PKL) and R (PKR) isoforms and the M1 (PKM1) and M2 (PKM2) isoforms. PKM1 is highly expressed in normal tissues, while high levels of PKM2 are found in proliferating cells [22, 23]. PKM2 can also regulate inflammasome activation in macrophages [24]. Here, for the first time, we studied the protein profiles of DM/PM muscle tissues through mass spectrometry analysis and investigated whether disrupted glycolysis promotes pyroptosis in muscle cells by activating the NLRP3 inflammasome.

Methods

Patients and controls

Detailed information on patients and normal controls (NCs) is presented in the Supplementary Materials, available at *Rheumatology* online. This study was approved by the ethics committee at Xiangya Hospital, Central South University of Changsha (Changsha, Hunan, China). All participants signed an informed consent form. The clinical characteristics of all patients were shown, and the detailed information for samples used in every experiment was listed in Supplementary Tables S1–S7, available at *Rheumatology* online (in these tables, clinical data were collected at the time of the first diagnosis excluding the patients after treatment, the latter we have marked 'Treatment after admission' and 'Therapeutic response').

Mass spectrometry and bioinformatics analysis

The muscle tissues from five DM and five PM patients and five NC subjects were obtained and sent to PTM BioLab (Hangzhou, China) for processing. Tandem mass tag labelling (Thermo Fisher Scientific, Waltham, MA, USA) was performed according to the manufacturer's protocol. The expression data were log2 transformed and analysed by a *t* test without error correction for the samples. Proteins with $P < 0.05$ and fold change > 1.5 were regarded as differentially expressed and selected for gene ontology (http://geneontology.org/) and reactome (https://reactome.org/) pathway enrichment by R clusterProfiler [25].

Cell culture and treatment with cytokines and inhibitors

Human skeletal muscle myoblasts were purchased from ScienCell (San Diego, CA, USA). Myotubes were obtained by replacing the growth medium with DMEM-F12 supplemented with 2% horse serum (Thermo Fisher Scientific, Waltham, MA, USA), penicillin and streptomycin for 5 days. Myotubes were exposed to IFNγ (PeproTech, RockyHill, NJ, USA) (100 ng/ml) for the indicated time in a 24-well plate (NEST Biotechnology, Wuxi, China). When required, the PKM2 inhibitor shikonin (Selleck, Houston, TX, USA) was added to the culture medium 2 h prior to IFNγ treatment.

RNA isolation and real-time quantitative PCR

Total RNA was extracted from muscle tissues and myotubes using TRIzol (Invitrogen Life Technologies, Waltham, MA, USA) according to the manufacturer's instructions. Real-time quantitative PCR (RT-PCR) was performed with TB Green PCR Master Mix (Takara). Each sample was run in duplicate on a 7500 Real-Time PCR system (Applied Biosystems, Waltham, MA, USA). Target gene expression relative to β-actin expression was measured by $2^{-\triangle\triangle CT}$. Detailed descriptions of all the primers are provided in Supplementary Table S8, available at *Rheumatology* online.

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Western blot analysis

A total of 80 μg of protein lysate extracted from muscle tissues or myotubes was loaded onto 12% or 15% gels for sodium dodecyl sulphf-polyacrylamide gel electrophoresis and transferred onto polyvinylidene difluoride membranes. The members were blocked with 5% non-fat dried milk in Tris-buffered saline containing 0.1% Tween-20 for 2 h and then incubated overnight at 4°C with anti-Vinculin (Sigma-Aldrich, St Louis, MO, USA), anti-NLRP3 (Cell Signalling Technology Inc, Danvers, MA, USA), anti-cleaved caspase-1 (Cell Signalling Technology Inc), anti-GSDMD (ABclonal, Wuhan, China) or anti-PKM2 antibodies (Cell Signalling Technology Inc). The next day, the membranes were incubated with secondary antibodies. ImageJ was used to quantify the intensity of the bands.

Haematoxylin and eosin and immunofluorescence (IF)

For histological analysis, the sections were stained with haematoxylin and eosin. For IF, the sections were permeabilized with 0.1% Triton X-100 for 15 min, blocked with 5% goat serum for 30 min, and incubated overnight at 4°C with goat anti-human NLRP3 (Abcam, Cambridge, UK, 1:200), rabbit anti-human caspase-1 (Abcam, 1:100) or rabbit anti-human PKM2 antibodies (Cell Signalling, 1:100). The sections were then incubated with secondary antibodies for 1 h at room temperature in the dark. The nuclei were stained with 4–6-diamidino-2-phenylindole. To evaluate cell IF, myotubes were fixed with 4% paraformaldehyde for 10 min, permeabilized with 0.5% Triton X-100 for 10 min and processed according to the tissue IF method.

FACS analysis

Pyroptosis was tested by using a FAM-FLICA Caspase-1 Detection kit (ImmunoChemistry Technologies, Bloomington, MN, USA) according to the manufacturer's instructions. After treatment with IFNγ and/or shikonin for 48 h, myotubes were stained with caspase-1 and propidium iodide (PI). The samples were run on a flow cytometer (BD Biosciences, San Diego, CA, USA) and analysed with FlowJo. Cells positive for both PI and caspase-1 were defined as pyroptotic cells.

ELISA and lactate dehydrogenase release

Plasma IL-1β concentrations were measured by a Human IL-1β/IL-1F2 Quantikine ELISA Kit (R&D Systems, Inc, Minneapolis, MN, USA). Lactate dehydrogenase (LDH) secretion in supernatants was measured with an LDH assay kit (Nanjingjiancheng, Nanjing, China) according to the manufacturer's instructions. The OD value was read at 450 nm with a microplate reader.

Statistical analysis

Normally distributed data were expressed as the means ± S.E.M., and non-normally distributed data were expressed as medians and interquartile range. Groups were compared by one-way ANOVA, Student's *t* test or the rank sum test. Categorical variables were presented as absolute values and percentages and were analysed by the χ^2 test or Fisher's exact test. A *P*-value <0.05 was considered statistically significant. All statistical analyses were performed by GraphPad Prism software (GraphPad Software, San Diego, CA, USA). The correlations between protein expression data and clinical data were assessed by Spearman correlation coefficients using the R ggcorrplot package.

Results

Glycolysis pathway proteins are upregulated in DM/PM by mass spectrometry analysis

Mass spectrometry analysis was performed in the muscle tissues of five DM and five PM patients and five NC subjects. Compared with the NC group, the DM group had 45 upregulated and 29 downregulated proteins, and the PM group had 49 upregulated and 31 downregulated proteins ($P<0.05$ and fold change>1.5). Among these dysregulated proteins, we identified six glycolysis related proteins in DM, and 14 in PM (Fig. 1A). Gene ontology annotation of biological processes revealed that multiple glycolysis processes, leucocyte migration and complement activation were enriched in both the DM and PM groups. Reactome pathway analysis suggested that glycolysis, programmed cell death and the NLRP3 inflammasome were dysregulated in both groups (Fig. 1B and C). Glycolysis-related genes such as hexokinase (HK) 2, phosphoglycerate kinase (PGK) 1, phosphoglycerate mutase 2 (PGAM2), enolase (ENO) 2, ENO3, glycerol phosphate dehydrogenase 1 (GPD1), PGK2, triosephosphate isomerase-1 (TPI1), glycosylphosphatidylinositol (GPI), lactate dehydrogenase A (LDHA), phosphoglucomutase 1 (PGM1), PKM, aldolase C (ALDOC), aldolase A (ALDOA), and glyceraldehyde-3-phosphate dehydrogenase (GAPDH) were increased in DM tissues compared with NC tissues, and further increased in PM tissues (Fig. 1D). Genes related the NLRP3 inflammasome and programmed cell death, such as mammalian target of rapamycin (MTOR), Pannexin1 (PANX1), NIMA-related kinase 7 (NEK7), high-mobility group box 1 (HMGB1), growth factor receptor-bound protein 2 (GRB2), Bcl-2-associated X protein (BAX), and eukaryotic translation initiation factor 5B (EIF5B) were also increased in most DM and PM patients compared with NC individuals, and the negative inflammasome regulatory gene, BCL2-associated transcription factor 1 (BCLAF1), was downregulated (Fig. 1E).

PKM2 is upregulated in skeletal muscle biopsies of DM/PM

We analysed the relationship between the altered glycolytic proteins and clinical data. However, no statistically significant relationships were observed (Supplementary

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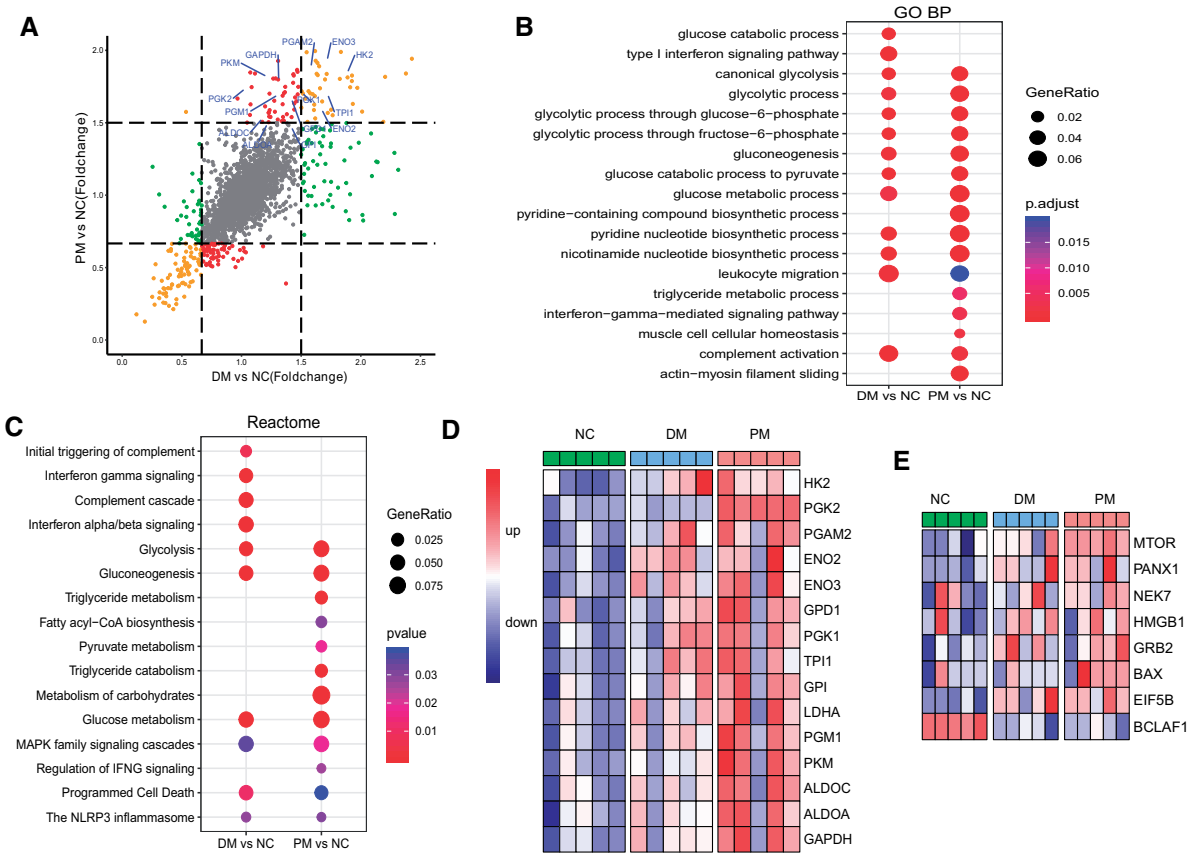
Fig. 1 Glycolysis pathway proteins are upregulated in DM/PM by mass spectrometry analysis



(**A**) Dysregulated proteins in the muscle tissues of DM and PM patients compared with NC. (**B**) GO analysis of the dysregulated proteins in DM and PM. (**C**) Reactome pathway analysis of the dysregulated pathways in DM and PM. (**D**) Heat map shows the glycolysis-related proteins in DM and PM. (**E**) Heat map shows the NLRP3 inflammasome and programmed cell death related proteins in DM and PM. GO: gene ontology; NLRP3: nucleotide-binding and oligomerization domain-like receptor family pyrin domain containing 3.

Fig. S2, available at *Rheumatology* online). Mass spectrometry analysis showed that PK protein levels were increased in DM/PM patients. We further detected PKM1 and PKM2 at the mRNA level by RT-PCR. PKM2 mRNA expression was significantly increased in the muscle tissues of PM patients, and showed an increasing tendency in DM patients. However, no significant difference was found for PKM1 (Fig. 2A). IF was performed to analyse the PKM2 expression pattern in muscle tissue sections. PKM2 levels were significantly higher in the sarcolemma of DM and PM samples than in the corresponding region of NC samples. In the PM group, PKM2 was also obviously increased in the cytoplasm of myofibres and in perivascular areas (Fig. 2B). Western blot analysis revealed that PKM2 was increased in all PM muscle tissues (9/9), and in 56% (5/9) of DM muscle tissues (Fig. 2C). However, we did not find a significant correlation between the levels of PKM2 and muscle enzymes (creatine kinase, CK, CK-MB, myoglobin, Mb and LDH) (Fig. 2D). PKM2 protein levels were associated with specific antibodies, and higher PKM2 levels were found in the anti-signal recognition particle (anti-SRP) and anti-Ro52 groups in DM/PM patients (Fig. 2E).

PKM2 colocalizes with NLRP3 in muscle tissue

We found that NLRP3 mRNA was increased in DM and PM muscle tissues compared with NC tissues, and NLRP3 mRNA levels increased with increasing levels of PKM2 (Fig. 3A). These results were confirmed by IF. PKM2 expression was mainly colocalized with NLRP3 in the sarcolemma and perimysium, but not in the cytoplasm in myofibres from DM/PM patients (Fig. 3B). Moreover, we confirmed PKM2 and NLRP3 protein levels in DM and PM muscles by western blot analysis. PKM2 and NLRP3 protein expression was increased in most PM muscle tissues and 1/3 of DM muscle tissues compared with NC muscle tissues. Furthermore, NLRP3 protein levels increased with increasing levels of PKM2 (Fig. 3C).

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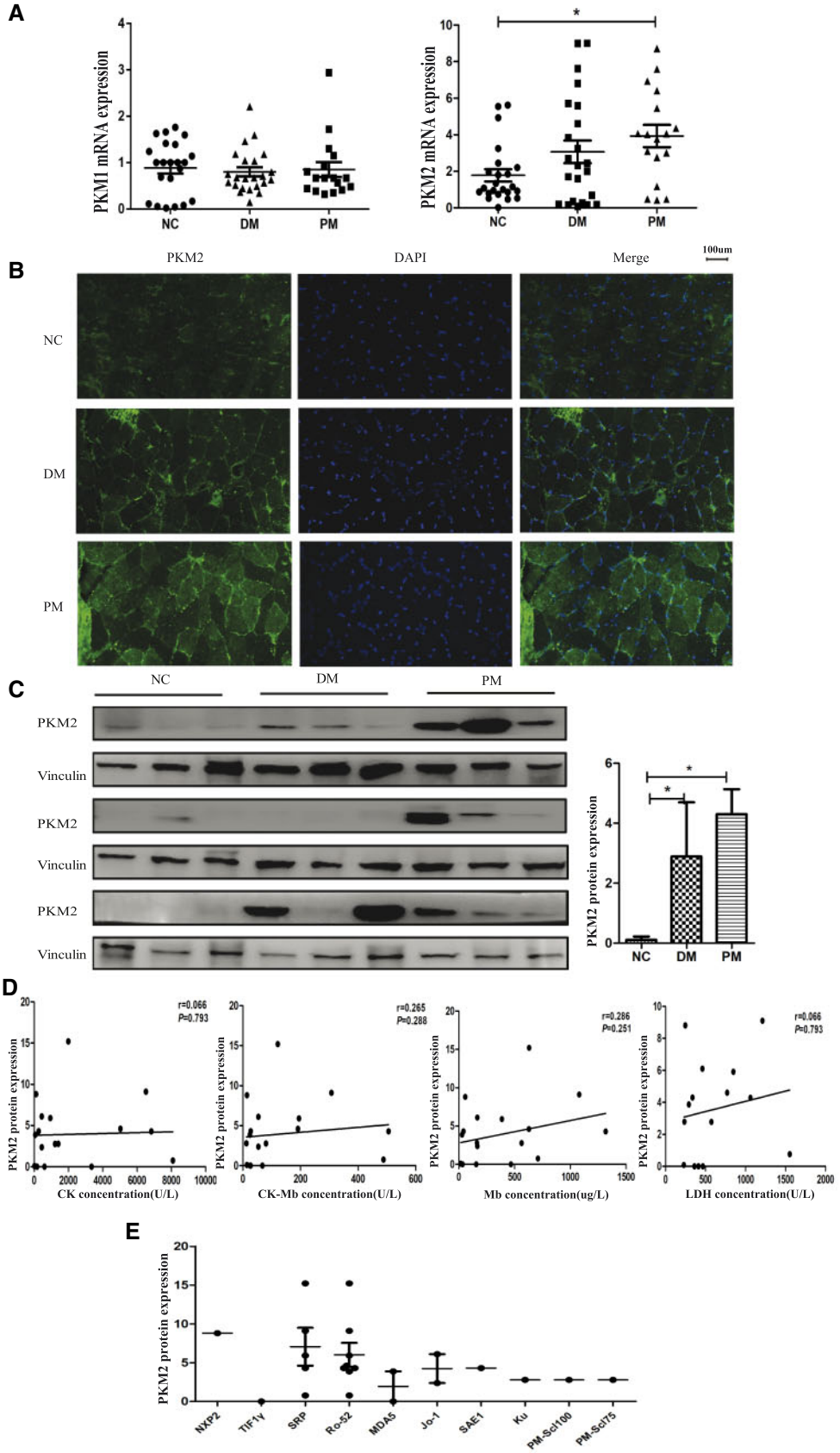
Fig. 2 PKM2 is upregulated in skeletal muscle biopsies of DM/PM



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NLRP3 inflammasome and pyroptosis pathways are activated in skeletal muscle biopsies of PM patients

Mass spectrum analysis suggested that programmed cell death and NLRP3 inflammasome pathways are involved in DM/PM. To confirm the levels of the NLRP3 inflammasome and pyroptosis in DM/PM patients, we performed RT-PCR to determine the mRNA levels of caspase-1, IL-1β, IL-18 and GSDMD. Caspase-1, IL-1β and IL-18 mRNA levels in muscle tissues were higher in DM/PM patients than in NC subjects, and GSDMD expression was only increased in PM patients (Fig. 4A). We further examined the expression pattern of the NLRP3 inflammasome in myofibres by IF and found that both caspase-1 and NLRP3 were predominantly localized in the sarcolemma of myofibres from DM and PM patients, and also expressed in the cytoplasm of scattered myofibres. Moreover, many myofibres in DM and PM patients that expressed NLRP3 were co-localized with caspase-1 (Fig. 4B). Cleaved caspase-1 and N-GSDMD protein levels were high in PM muscles and slightly elevated in DM muscles (Fig. 4C). Then, we analysed the plasma IL-1β concentration and found that IL-1β levels were increased in PM samples, but not in DM samples, compared with NC samples (Fig. 4D). IL-1β levels were associated with specific antibodies in DM/PM patients. Specifically, IL-1β expression was higher in patients expressing the anti-SRP antibody than in patients expressing the anti-transcriptional intermediary factor 1 γ (TIF1γ), anti-PM-Scl75 and anti-Mi-2β antibodies. IL-1β levels in patients expressing the anti-TIF1γ antibody were lower than those in patients expressing the anti-Ro-52, anti-melanoma differentiation-associated gene 5 (MDA5), anti-PL-7, anti-PM-Scl100 and anti-PM-Scl75 antibodies (Fig. 4E). IL-1β levels also positively correlated with myocardial enzymes (CK, CK-MB, Mb, LDH, alanine transaminase and aspartate transaminase) (Fig. 4F).

PKM2 inhibitor blocks IFN-induced NLRP3 activation and pyroptosis in myotubes

An activated NLRP3 inflammasome pathway was reported in human myotubes under IFNγ-induced inflammatory conditions [26]. To explore the relationship between PKM2 and the NLRP3 inflammasome and pyroptosis, we stimulated myotubes with IFNγ. After 48 h of exposure to IFNγ (100 ng/ml), the mRNA levels of NLRP3, caspase-1, IL-1β and IL-18 were significantly upregulated in myotubes (Fig. 5A). The protein levels of NLRP3, cleaved caspase-1 and N-GSDMD were also increased after IFNγ stimulation (Fig. 5D). To investigate the role of PKM2 in NLRP3 inflammasome activation and pyroptosis, myotubes were treated with 1 μM shikonin, a potent PKM2 inhibitor, for 2 h prior to IFNγ exposure for 48 h. Shikonin inhibited IFNγ-induced NLRP3, caspase-1, IL-1β and GSDMD mRNA expression in myotubes, but had no effect on IL-18 mRNA expression (Fig. 5B). Similarly, western blot analysis showed that shikonin reduced cleaved caspase-1 and N-GSDMD protein levels in myotubes stimulated with IFNγ for 48 h (Fig. 5D). IF analysis of IFNγ-treated myotubes confirmed that NLRP3 expression was inhibited by shikonin (Fig. 5C). A kit including PI and FAM-YVAD-FMK, a caspase-1-specific marker, was used to examine myotube pyroptosis [27]. Under normal cell culture conditions, few myotubes underwent pyroptosis. However, after treatment with IFNγ (100 ng/ml) for 48 h, myotubes showed a remarkable increase in pyroptosis, and shikonin inhibited the development of pyroptosis (Fig. 5E). In addition, shikonin inhibited the increased LDH levels in the culture supernatant of IFNγ-stimulated myotubes (Fig. 5F).

Discussion

Metabolic reprogramming has emerged as a crucial player in cancer progression as well as autoimmune diseases. Mitochondrial abnormalities and metabolic disturbances have also been observed in IIM patients. The upregulation of succinate dehydrogenase, the key enzyme in oxidative phosphorylation, and the downregulation of cytochrome c oxidase were observed in the impaired muscle fibres of DM and IBM patients [27, 28]. Adenosine triphosphate levels in the muscles of myopathic DM patients were lower than those in NC subjects or amyopathic DM [29]. A multiplex assay of muscle proteins involved in energy metabolism suggested a repressed glycolytic metabolism in IBM and upregulated PKM2 expression in DM [30]. Here, we showed that multiple glycolysis-related processes, the NLRP3 inflammasome and the programmed cell death pathway were dysregulated in the muscle tissues of DM and PM patients. Multiple proteins in the glycolysis pathway were significantly increased in PM patients and mildly increased in DM patients. Previous studies indicated that disrupted metabolic homeostasis might lead to irreversible structural damage to muscle fibres [1].

Fig. 2 Continued

(**A**) The mRNA levels of PKM1 and PKM2 in DM and PM muscle tissues. (**B**) Expression of PKM2 (green) was increased in the sarcolemma of DM (patient 40) and PM (patient 46) samples by IF. In PM group, PKM2 was also obviously elevated in the cytoplasm of myofibre and perivascular areas. (**C**) Western blot and semi-quantitative analysis of PKM2 in total protein lysate of NC, DM and PM patients muscle biopsies (patients' samples from left to right and then from top to bottom: 13, 15, 21, 44–46, 22, 23, 34, 47, 51, 52, 35–37, 53, 57, 58). (**D**) A lack of correlation was found between the protein level of PKM2 and muscle enzymes. (**E**) PKM2 levels were associated with specific antibodies in DM/PM patients. *$P<0.05$. Data are expressed as means ± s.e.m. PKM2: pyruvate kinase isozyme M2.

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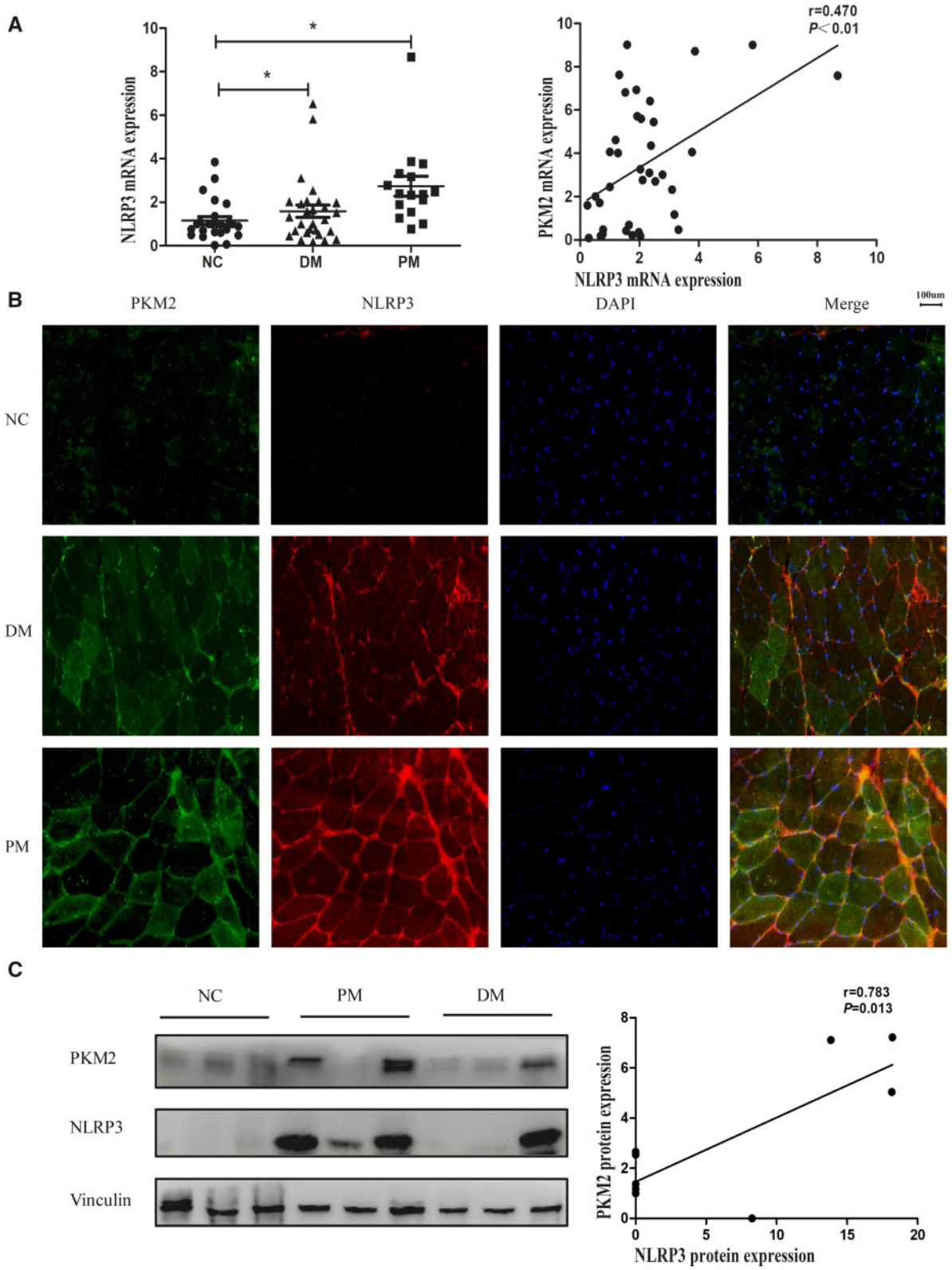
Fig. 3 PKM2 co-locates with NLRP3 in muscle tissue



(**A**) The mRNA expression of NLRP3 was increased and had a positive relationship with PKM2 expression in muscle tissue of DM and PM. (**B**) Co-staining of PKM2 (green) and NLRP3 (red) in muscle tissues of DM (patient 42), PM (patient 50) and NC. (**C**) Western blot and semi-quantitative analysis of PKM2 and NLRP3 in total protein lysate of NC, DM and PM patients' muscle biopsies (patients' samples from left to right: 59–61, 38–40). The protein level of NLRP3 had a positive relationship with PKM2 (r=0.783, P =0.013). Data are expressed as means ± S.E.M. NLRP3: nucleotide-binding and oligomerization domain-like receptor family pyrin domain containing 3; PKM2: pyruvate kinase isozyme M2.

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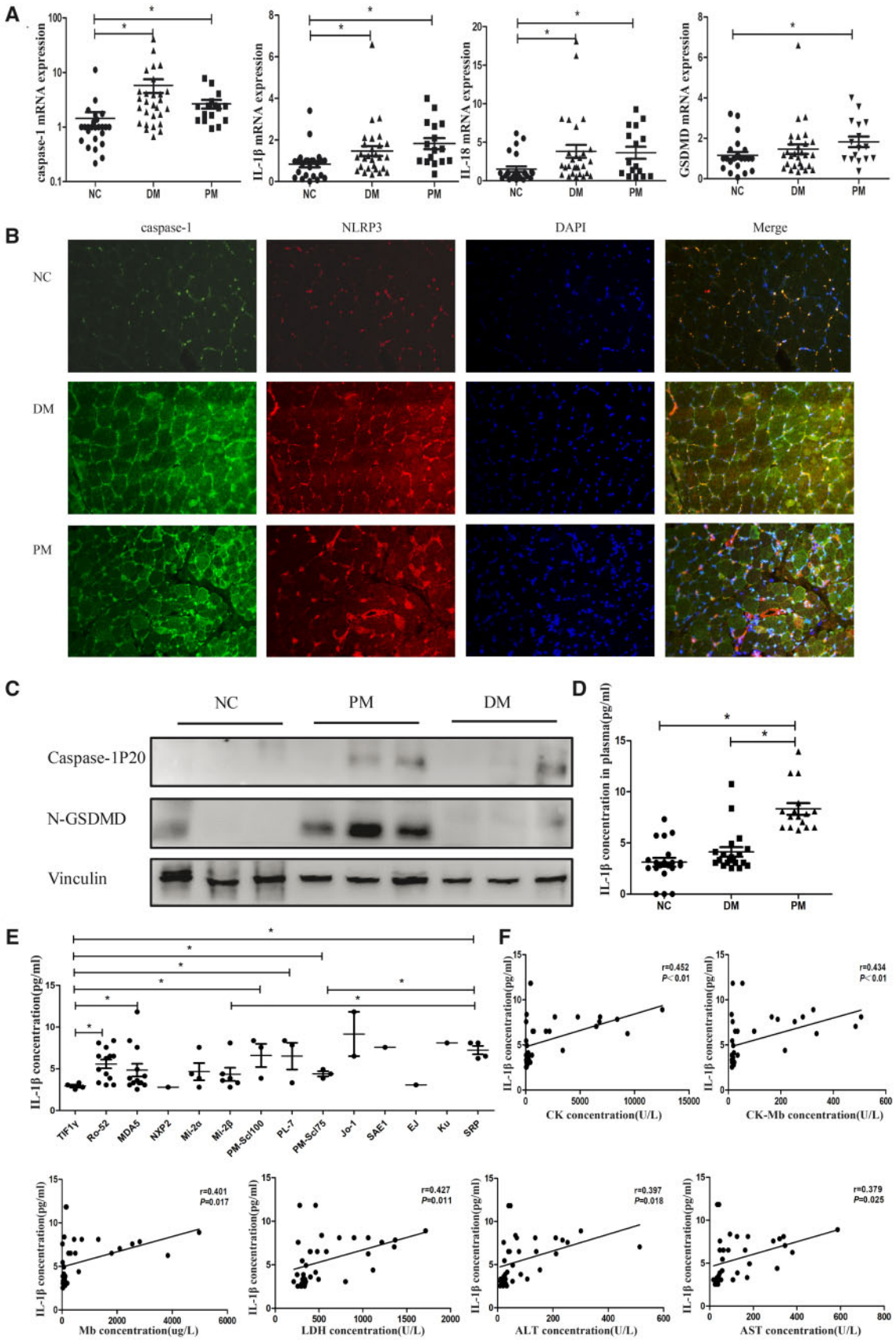
Fig. 4 The NLRP3 inflammasome and pyroptosis pathways are activated in skeletal muscle biopsies of PM



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In addition to cancer cells and normal proliferating cells, some activated immune cells also express PKM2, wherein it plays a central role in cell proliferation and survival by regulating endogenous allosteric effectors or intracellular signalling pathways or through noncanonical nuclear functions [31, 32]. Elevated PKM2 levels have been reported in multiple autoimmune diseases, such as rheumatoid arthritis, and inflammatory bowel disease, as well as in sepsis [24, 33, 34]. Here, we revealed that PKM2 was upregulated in the muscle fibres of DM patients and highly upregulated in PM patients compared with NC individuals, and that PKM2 expression levels had a positive relationship with NLRP3 levels.

The discovery of novel autoantibodies has major implications for distinguishing the entity of IIMs [35]. The anti-SRP autoantibody, which is directed against components of SRP, is associated with necrotizing myopathy. Anti-SRP autoantibody-positive IIM patients often have severe muscle weakness and are resistant to glucocorticoid therapy [36]. Interestingly, we found high PKM2 and IL-1β levels in patients with anti-SRP autoantibodies. Moreover, IL-1β levels positively correlated with myocardial enzymes levels. These results suggest the important roles of PKM2 and IL-1β in aggressive muscle damage. Decreased IL-1β levels were observed in patients with anti-TIF1γ autoantibodies, which are associated with cancer in adult DM patients and with skin rashes in all DM subtypes, but not with muscle damage [37].

Interestingly, some IIM patients show severe muscle damage without significant inflammatory cell infiltration, which is inconsistent with the general notion that autoimmune and inflammatory processes result in muscle fibre necrosis. A lack of relationship between muscle weakness and inflammatory infiltrates has been observed in IIMs [5, 38]. Increased class I major histocompatibility complex (MHC-I) expression was found in muscle tissues from DM/PM patients with persisting muscle weakness independent of adjacent inflammatory cell infiltration [5]. In a mouse model of inflammatory myopathy, metabolic disturbances and muscle weakness were detected prior to inflammatory cell infiltration [39]. Here, we revealed multiple dysregulated glycolysis-related proteins, especially PKM2, in muscle tissues of DM/PM patients. These molecular changes may be associated with the induction of inflammatory processes and disrupted muscular homeostasis, which subsequently lead to muscle cell death.

Moreover, we revealed the upregulation of N-GSDMD, the executioner of pyroptosis, especially in the muscle fibres of PM patients. This finding indicates that pyroptotic cell death may occur in muscle cells of patients with IIMs. Myofibre necrosis is one pathological feature of affected muscle tissues in IIMs [40]. It is generally thought that CD8+, CD4+/CD28-null and CD8+/CD28-null T cells can recognize MHC-I molecules on non-necrotic muscle fibres and induce muscle fibre injury through a perforin- and secretion-dependent mechanism [41, 42]. Pyroptosis is a kind of gasdermin-mediated programmed necrotic cell death [43]. In contrast to other types of cell death, such as necrosis and apoptosis, pyroptosis causes rupture of the cell membrane, releases pro-inflammatory cytokines, and damage-associated molecular patterns, and results in the marked amplification of the inflammatory process [44]. The new finding of muscle cell pyroptosis may play a more important role in muscle damage associated with IIMs compared with other cell death types. It also suggests that muscle cells actively participate in the immune response and myofibre damage, which might partly explain the phenomenon of muscular necrosis and inflammation without inflammatory cells infiltrations in IIMs [4].

Recent studies have identified roles for glycolysis, lipid metabolism and mitochondrial reactive oxygen species (ROS) in regulation of the NLRP3 inflammasome [45, 46]. In macrophages, the glycolytic enzymes HK1 and PKM2 have been confirmed to activate the NLRP3 inflammasome [24, 47]. However, whether PKM2 can induce NLRP3-dependent pyroptosis is still unclear. Here, we first showed that shikonin, a PKM2 inhibitor, can prevent the pyroptosis of muscle cells by inhibiting the NLRP3 inflammasome activation. Shikonin, a Chinese medicine isolated from the herb *Lithospermum erythrorhizon*, has been shown to have anticancer and anti-inflammatory properties [48, 49]. Shikonin may have potential as an IIMs therapy. However, the mechanism through which PKM2 regulates NLRP3 inflammasome activation and pyroptosis in muscle cells needs to be further studied.

In summary, we investigated the expression levels of the NLRP3 inflammasome, and components of glycolysis and pyroptosis and the relationship among these factors in IIMs. We suggest a role for PKM2 in promoting the pyroptosis of muscle cells via the NLRP3 inflammasome. Our research broadens the study of glycolytic

Fig. 4 Continued

(**A**) The mRNA expression of caspase-1, IL-1β, IL-18 was elevated in muscle tissue of DM and PM compared with NC. The mRNA expression of GSDMD was only elevated in muscle tissue of PM. (**B**) Co-staining of caspase-1 (green) and NLRP3 (red) in muscle tissues of DM (patient 38), PM (patient 54) and NC. (**C**) Western blot analysis of caspase-1p20 and N-GSDMD in total protein lysate of NC, DM and PM patients muscle biopsies (patients' samples from left to right: 44–46, 13, 15, 21). (**D**) IL-1β concentrations were increased in the plasma of PM patients. (**E**) IL-1β levels were associated with specific antibodies in DM/PM patients. (**F**) IL-1β concentrations had a positive correlation with myocardial enzymes. *$P<0.05$. Data are expressed as means ± s.e.m. GSDMD: gasdermin D.

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Fig. 5 PKM2 inhibitor blocks IFNγ-induced NLRP3 activation and pyroptosis in myotubes

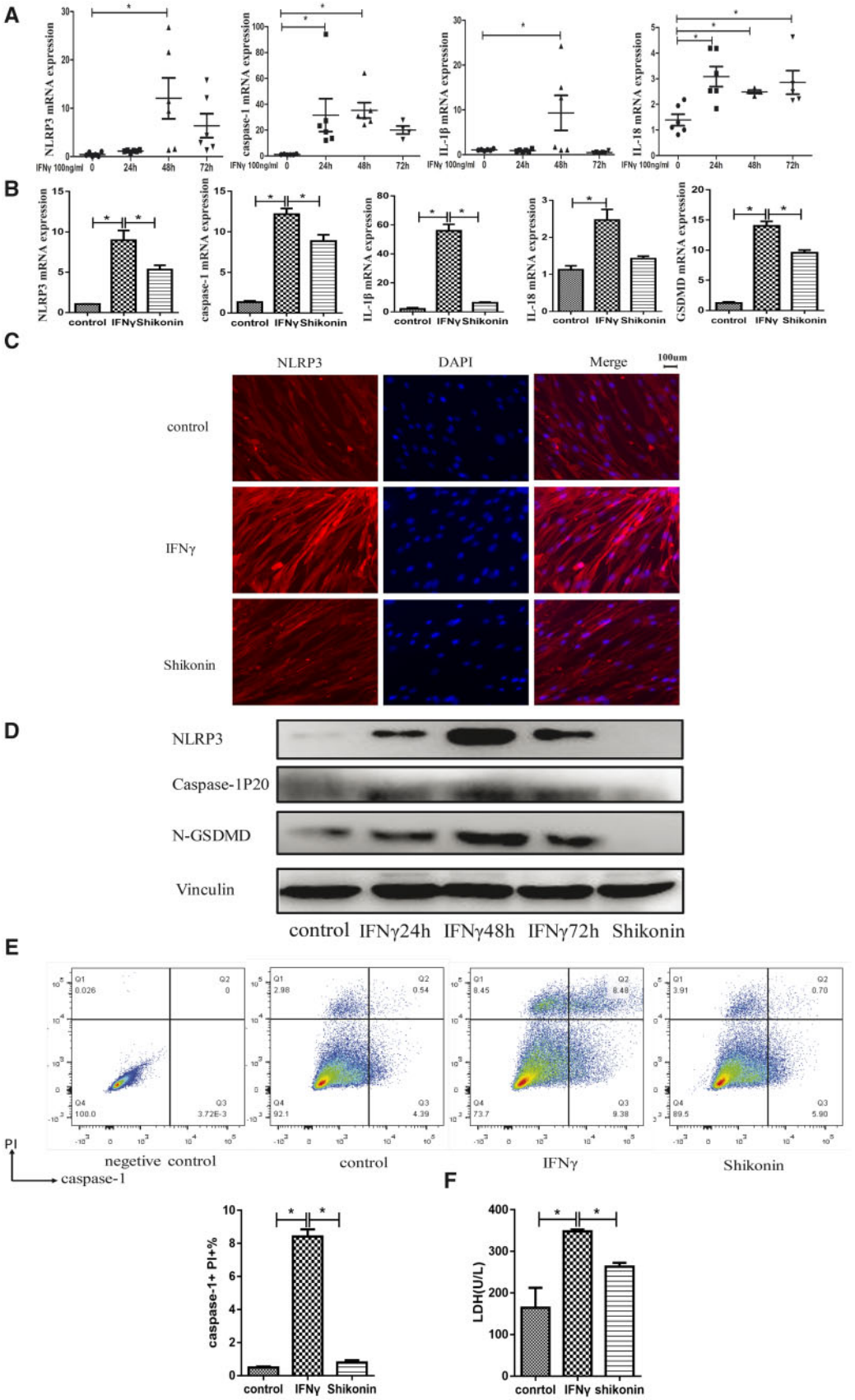

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Fig. 5 Continued
(**A**) The mRNA expression of NLRP3, caspase-1, IL-1β and IL-18 in myotubes after incubating with 100 ng/ml IFNγ for 24, 48 and 72 h, respectively. (**B**) The mRNA expression of NLRP3, caspase-1, IL-1β, IL-18 and GSDMD in myotubes after treatment with 1 μm PKM2 inhibitor shikonin for 2 h prior to the induction by IFNγ for 48 h. (**C**) IF staining of NLRP3 (red) in myotubes incubated with IFNγ for 48 h treatment with or without shikonin (1μm). (**D**) Western blot analysis of NLRP3, caspase-1p20 and N-GSDMD in total protein lysate of myotubes after incubating with 100 ng/ml IFNγ for 24 h, 48 h and 72 h. (**E**) Flow cytometry dot blots and quantification of caspase-1 and PI double-positive cells were shown in each group. (**F**) Inhibition of PKM2 by shikonin (1 μm) significantly inhibited IFNγ induced LDH production in myotubes. $^*P < 0.05$. Data are expressed as means ± s.e.m. of three independent experiments. GSDMD: gasdermin D; NLRP3: nucleotide-binding and oligomerization domain-like receptor family pyrin domain containing 3; PKM2: pyruvate kinase isozyme M2.

metabolism abnormalities in IIMs and may provide a promising target for the treatment of IIMs.

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Supplementary data

Supplementary data are available at *Rheumatology* online.

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